

Source-Bounded Exact Recovery over Docker’s Logs API

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Abstract

Reliable container-log acquisition faces a specific problem: Docker can retain records that a log follower misses before attachment or during collector downtime. We study recovery over Docker’s supported Logs API and define *source-bounded exactness*: every retained, distinguishable source record eventually appears exactly once in durable collector output. Our method uses a generation-aware multiset oracle that separates source truncation from collector omission and exposes simultaneous loss and replay that count arithmetic hides. Applying the oracle to LogDeck uncovered a start-to-attachment race in which an initial backfill ended before a prefix was emitted and the live stream attached afterward. A one-record attachment overlap, finite Docker-API reconciliation, and exact insertion closed this tested boundary.

We compare the fixed revision with unmodified Grafana Alloy 1.18.0, which uses the same supported Docker Logs API and persists read positions. Across 120 Linux/arm64 collector-runs (two collectors, six faults, ten trials, 5,000 records), LogDeck was exact in 60/60 and Alloy in 20/60. Alloy was exact for guarded startup and process pause, but missed rapid-exit containers and retained only the pre-fault prefix after collector restart, crash, and a restart reusing the same physical container identifier (ID); one same-ID trial also replayed 1,479 prefix records. Both collectors were exact in 20/20 additional collector-runs across an isolated live-restore Docker daemon restart. After verified source removal, both retained only the already-acquired prefix in 20/20 runs, marking the recovery horizon. A stateful Fluent Bit file-tail control matched LogDeck across restart and crash but required Docker’s private storage. Exactness assumes distinct observable tuples of physical generation, timestamp, stream, and bytes; 200,000 byte-identical records across the `json-file` and `local` drivers produced no observed collisions. These results show that lifecycle reacquisition, not a persisted position alone, determines exact recovery within the retained-source horizon; the collision experiment is evidence rather than a universal guarantee.

1 Introduction

An application writing a line to standard output does not imply that an external collector durably acquired it. Between the

write and a queryable store are a logging driver, finite retained history, Docker’s Logs API, a followed HTTP response, collector buffers, and durable storage. These boundaries do not fail in the same way.

Durable logs, checkpoints, and replay are established ideas. Chukwa places recovery state near local sources and retransmits retained data [13]; Kafka provides a durable append-only log with consumer-managed positions [10]; and Mill-Wheel combines persistent state with fault-tolerant stream processing [1]. The open engineering question here is narrower: what recovery contract can a collector provide when its source is Docker’s supported Logs API rather than a durable broker or direct access to the daemon’s files?

The answer depends on more than whether a collector stores an offset. A collector must first discover the physical container, attach to a stream, recognize lifecycle changes, and enumerate retained history after a fault. Those actions interact with a source whose contents and lifetime are controlled by Docker. A stopped container can still have readable history; a removed container does not.

Measuring this boundary is also subtle. Suppose a collector misses one identity and repeats another. Its output count is unchanged. Similarly, reusing sequence numbers after a same-ID restart can make two physical generations appear to be one. Existing collector benchmarks already study throughput, resource use, rotation, and missing records [18]. Our focus is therefore not a generic collector ranking, but a duplicate-aware oracle and a lifecycle fault model that can localize why acquisition is incomplete.

This study asks three questions:

- RQ1** Can a Docker API follower converge to exactly one durable copy of every retained, distinguishable source record?
- RQ2** Which attachment, collector, daemon, and container lifecycle boundaries cause omission or replay?
- RQ3** Which mechanism is necessary for recovery, and where does the source lifetime make recovery impossible?

The central result is not that one collector wins every condition. The Alloy control succeeds at guarded startup, process pause, and live-restore daemon restart. It fails when the tested schedule requires reacquisition after the collector process or container lifecycle changes. A private-file Fluent Bit

control also recovers exactly across collector restart and crash. These contrasts place the contribution at the combination of supported interface, lifecycle reconciliation, and exact oracle.

This paper makes four contributions:

1. a source-bounded acquisition contract and a generation-aware multiset oracle that separately reports source truncation, omission, replay, unexpected identities, and malformed output;
2. a Docker lifecycle suite spanning attachment, ephemeral exit, collector restart/crash/pause, same-ID restart, daemon restart, source rotation, and container removal;
3. a concrete account of a start-to-attachment race found by the harness, together with a finite API-reconciliation design and controlled ablations of reconciliation and exact insertion; and
4. comparative evidence from a supported-API baseline, a private-file baseline, and two Docker logging drivers that identifies both recoverable and unrecoverable boundaries.

2 Model and Oracle

2.1 Log path and source horizon

We model the acquisition path as

stdout/stderr $\rightarrow$ driver history $\rightarrow$ Docker Logs API
 $\rightarrow$ collector buffer $\rightarrow$ durable store.

The selected logging driver and its retention configuration determine whether historical logs remain readable [3]. No downstream algorithm can recover a record after the source deletes the only retained copy. The contract must therefore name the observation horizon explicitly.

2.2 Source-bounded exactness

Every generated line contains a run token, a logical generation, and a sequence number. Let G be the complete set of generated benchmark identities. After the workload, the harness materializes Docker’s retained history; let $S \subseteq G$ be the set visible at that observation point. Let C be the collector output multiset, and let $m_C(r)$ be the observed multiplicity of identity r .

The primary contract is

$$\forall r \in S : m_C(r) = 1.$$

The harness reports four disjoint questions rather than a single count:

- $G \setminus S$: generated but absent from retained source;

- $\{r \in S : m_C(r) = 0\}$: source-retained but collector-missing;
- $\sum_{r \in G} \max(0, m_C(r) - 1)$: excess multiplicity; and
- $C \setminus G$: unexpected identities.

Malformed benchmark lines are reported separately. The same-ID scenario increments the logical generation even though Docker reuses the physical container ID. The oracle materializes all identities; it never infers loss from a maximum sequence number or a total.

2.3 Observable identity assumption

The implementation does not receive the benchmark identity as privileged metadata. It deduplicates the Docker-visible tuple

(physical generation, timestamp, stream, raw bytes).

Two legitimate records with the same four fields are observationally indistinguishable at this interface. In the main workload, raw bytes contain the benchmark generation and sequence, so every expected identity is distinguishable. A separate stress test removes that convenience by emitting byte-identical records and measuring whether Docker supplies distinct timestamps. This tests an important ordinary case but cannot prove that a collision is impossible on every driver, engine, or platform.

3 Recovery Design and the Attachment Race

3.1 Finite reconciliation

LogDeck keeps the live path non-blocking. Followers feed bounded in-memory queues, and a slow consumer cannot stall acquisition indefinitely. Queue overwrite and lifecycle transitions produce reconciliation hints. Per-stream watermarks record durable progress, while a startup snapshot prevents a newly advanced live watermark from causing older retained history to be skipped.

A reconciliation pass rereads a finite Docker history interval with overlap before the durable watermark or earliest gap. Inserts are idempotent under the observable identity key. Gap hints are versioned: a pass may clear only the version it observed, so a newer gap cannot be erased by stale recovery work. Hints coalesce into an immediate wake, and periodic inventory remains a safety net.

This is conventional at-least-once reread plus deduplication. The Docker-specific issue is deciding *which physical containers* and *which lifecycle intervals* must be reread through an API whose source can outlive a running process but not container removal.

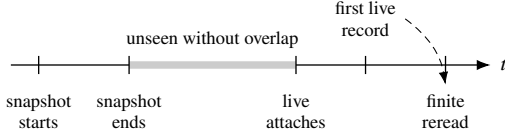


Figure 1: The start-to-attachment race and repair. A first-record attachment overlap triggers a finite reread; exact insertion removes the repeated boundary record.

3.2 A race exposed by the oracle

The guarded runner found a schedule absent from the original randomized matrix. At revision b06a1ed, an initial finite backfill completed before the workload emitted its prefix. The Tail: 0 live response attached after that prefix. Because the producer then waited at a gate, no later record arrived to trigger another recovery pass. The database contained the discovered target and completed-backfill state but no target rows or watermark.

Focused failing tests reproduced this ordering. The repair asks each persistent subscription for one retained record when opening the followed stream. The first delivered record proves attachment and triggers one finite zero-point reread. The reread overlaps the live boundary, and exact insertion absorbs the intentional repeated record. Retried attachments apply the rule once per attachment. Figure 1 illustrates the schedule and repair.

3.3 Bounded design claim

Recovery is intended to converge when (1) the relevant record remains readable from finite Docker history, (2) the collector can identify its physical generation, (3) reconciliation eventually runs to completion, and (4) distinct records have distinct observable identities. The evaluation tests these assumptions and necessary mechanisms; it does not claim an unrestricted exactly-once channel.

4 Methodology

4.1 Black-box harness

The standalone Go harness imports no LogDeck package. It controls collectors and producers through Docker and public HTTP endpoints. Every collector, scenario, and trial receives a unique run token and target label. The expected identity set is created before execution. After the workload, the harness reads Docker history, snapshots collector output until exactness remains stable for one second or an incomplete result plateaus, and writes a schema-versioned JSON report. Raw sources and collector artifacts can be retained independently.

Fault boundaries are selected reproducibly between 20% and 80% of the 5,000-record workload. The seed depends on scenario and trial but not collector, so both collectors receive

the same ten boundaries. A *collector-run* is one collector executing one scenario once. All main-table denominators use this unit.

4.2 Collectors

LogDeck. The evaluated image is built from revision 58b3e4b and pinned by digest prefix bd6b4630. It includes the attachment repair. Research ablation modes disable only reconciliation or exact insertion in the same instrumented revision.

Grafana Alloy supported-API control. We run unmodified Grafana Alloy 1.18.0, digest prefix 491b0578. Its `discovery.docker` component supplies labeled targets to `loki.source.docker`, which reads the Docker Logs API and stores read offsets in a persistent positions file [9]. A `loki.echo` receiver makes records observable to the harness. The configuration uses a 100 ms discovery and source refresh interval. Before guarded faults, the harness queries Alloy’s component debug endpoint and releases the producer only after the expected target reports a running reader. Thus startup and pre-fault failures cannot be attributed to releasing the workload before attachment.

Fluent Bit private-file control. Pinned Fluent Bit 5.0.9 tails Docker’s `json-file` records with a persistent offset database. A Linux-process variant survives a real process stop or SIGKILL. This is a useful stateful comparison, but it reads the daemon’s private log storage. Docker explicitly warns that these files are designed for exclusive daemon access [5]; the privilege and coupling difference is therefore part of the result.

4.3 Faults

Table 1 defines the principal schedules. Guarded scenarios confirm attachment before their measured prefix. The rapid-exit scenario is intentionally unguarded because discovery of an ephemeral producer is the fault under test.

The daemon fault runs the harness inside a privileged, isolated Docker-in-Docker engine so the ordinary Ubuntu daemon is untouched. A supervisor restarts only the inner `dockerd`. Live restore is enabled and verified before each matrix; Docker documents that this mode keeps containers running while the daemon is unavailable [6]. The producer begins its suffix during the outage. It may block if Docker’s first-in, first-out (FIFO) logging buffer fills, an explicit limitation of live restore.

For removal, the harness stops the collector after the prefix, releases and waits for the complete producer, materializes and verifies all 5,000 source identities, deletes the container, and restarts the collector. The report marks this pre-removal oracle explicitly; the post-removal source horizon is empty.

Table 1: Principal fault schedules. Prefix and suffix refer to the seeded fault boundary.

Fault	Schedule
Startup	Emit prefix; start collector; confirm attachment; emit suffix.
Rapid exit	Start collector; producer emits all records and exits immediately.
Restart	Acquire prefix; stop collector; emit suffix and exit; restart collector.
Crash	Acquire prefix; SIGKILL collector; emit suffix and exit; restart.
Pause	Acquire prefix; pause process; emit suffix and exit; resume.
Same ID	Acquire generation 1; restart the same physical container for generation 2.
Daemon	Begin suffix while an isolated live-restore daemon is unavailable.
Removal	Verify full source, delete container, then restart collector.

4.4 Environments

The supported-API and removal matrices run inside the default Ubuntu OrbStack machine on Linux/arm64 with eight CPUs and Docker Engine 29.1.3. The machine has a separate Linux userspace and Docker daemon but shares the macOS host’s OrbStack kernel; it is not independent hardware. The daemon matrix uses the same kernel with an isolated Docker 29.7.0 live-restore engine.

Earlier fixed-revision work was replicated on an independent Ubuntu/arm64 cloud host with Linux 5.15 and an isolated Docker 24.0.9 daemon. That replication and the independent Fluent Bit process control support the mechanism and private-file results, but the new Alloy comparison is reported only for the Docker 29.1.3 Ubuntu-machine environment. We keep this distinction explicit rather than treating containerized Linux as a second physical host.

5 Results

5.1 The supported-API comparison localizes lifecycle recovery

Table 2 reports 120 collector-runs: two collectors, six scenarios, and ten trials per cell. There were no operational errors. LogDeck was source-bounded exact in all 60 of its collector-runs. Alloy was exact in 20/60: all startup and pause trials.

All ten Alloy rapid-exit runs acquired zero of 5,000 records even though the post-run Docker oracle retained all 5,000. In restart and crash, Alloy’s final unique count equaled the randomized pre-fault boundary in every run; the entire suffix was absent. The same was true after the same physical container ID restarted for generation 2. Since Alloy succeeds when the existing reader is paused and when a guarded startup

Table 2: Supported Docker-API comparison, 5,000 records per collector-run. Each cell is exact collector-runs out of ten.

Fault	LogDeck	Alloy	Alloy diagnostic
Startup	10/10	10/10	known-target exact
Rapid exit	10/10	0/10	none acquired
Restart	10/10	0/10	pre-fault only
Crash	10/10	0/10	pre-fault only
Pause	10/10	10/10	reader catch-up
Same ID	10/10	0/10	generation 1 only
Total	60/60	20/60	

target is known, this pattern is consistent with a lifecycle reacquisition or rediscovery boundary rather than a general inability to read historical records.

The multiset oracle found more than omission. Same-ID trial 6 retained 3,647 unique generation-1 identities, missed all 1,353 generation-2 identities, and replayed g1:1--1479. Its final total was 5,126. A total-only method would report 126 extra records and conceal both the missing suffix and 1,479 replays.

5.2 Daemon restart is a shared success

The isolated live-restore matrix contains 20 collector-runs: ten per collector, with the same ten randomized boundaries. Both LogDeck and Alloy were generated-set exact in 10/10, with no duplicate, unexpected, or malformed identity and no operational error. The result rules out an overly broad claim that Alloy cannot recover from Docker API disruption. When the collector process and its in-memory state survive the daemon outage, the tested configuration catches up exactly.

5.3 Container removal marks the recovery horizon

Immediately before each deletion, the independent oracle and Docker history both contained all 5,000 generated identities. After deletion and collector restart, both collectors ended with exactly the randomized pre-removal prefix in all ten trials. Across 20 collector-runs, final unique count equaled the fault boundary every time and duplicates were zero.

This is not a violation of source-bounded exactness: after removal, Docker no longer exposes the suffix. It is direct evidence for the contract’s limit. A collector can preserve what it already acquired, but API reconciliation cannot recover a source object that no longer exists.

5.4 The harness found and closed a real race

Before the attachment repair, the guarded runner repeatedly reached a state with a completed initial backfill, an attached live stream, and no acquired prefix. Focused tests forced that schedule. After the one-record overlap and finite reread were

Table 3: Fixed-revision ablations, ten collector-runs per cell. Ranges report missing or excess multiplicity per run.

Mechanism/fault	Complete	Disabled
Reconcile/start	exact 10/10	1,458–3,498 missing
Reconcile/exit	exact 10/10	671–1,158 missing
Exact insert/start	exact 10/10	5,001–10,001 extra
Exact insert/restart	exact 10/10	1,805–6,464 extra

Table 4: Independent collector-process control, ten collector-runs per cell.

Fault	LogDeck API	Fluent Bit private file
Graceful restart	10/10	10/10
SIGKILL	10/10	10/10

added at revision 58b3e4b, both the local and independent fixed-revision matrices completed without the pre-fault timeout. In the new API comparison, guarded startup was exact in 20/20 collector-runs, including Alloy, confirming that the harness does not manufacture a startup failure for the comparator.

This before/after sequence is a first-class result: the oracle and guarded schedule exposed an acquisition hole that an unguarded burst benchmark had not, and the repaired schedule remained exact across the evaluated environments.

5.5 Ablations identify two necessary mechanisms

Table 3 holds the revision and attachment overlap fixed while disabling one mechanism. There were 80 collector-runs across the two paired matrices. Every complete-design control was exact; every disabled condition was inexact.

Without reconciliation, startup missed exactly one fewer record than the seeded prefix in every trial, consistent with the preserved one-record attachment overlap. Without exact insertion, the collector acquired every unique identity but retained repeated copies from overlapping rereads. These results establish necessity under the tested faults, not sufficiency under every Docker failure.

5.6 A private-file baseline changes the interface tradeoff

On the independent Linux process control, LogDeck and Fluent Bit were both generated-set exact in all 40 collector-runs: ten graceful restarts and ten SIGKILL trials per collector (Table 4). A broader 100-run supplemental matrix also found both exact across startup, rapid exit, pause, same-ID restart, and source-horizon conditions.

The result prevents a claim that LogDeck is uniquely exact. Stateful file offsets are an effective recovery primitive

Table 5: Byte-identical identity stress, ten collector-runs per driver.

Driver	Oracle/API	Tuple collisions	Exact
json-file	100,000/100,000	0	10/10
local	100,000/100,000	0	10/10

here. The distinction is deployment: LogDeck uses Docker’s supported API; this Fluent Bit control requires the isolated daemon’s private json-file data root.

5.7 Observable identity survives two tested drivers

The identity stress emits the same raw stdout bytes 10,000 times per trial and writes an independent bind-mounted oracle before each emission. The harness reads Docker history with timestamps and counts collisions in the visible tuple. Table 5 reports 20 trials and 200,000 records.

Every Docker-visible record had a distinct nanosecond timestamp, and LogDeck preserved exactly 100,000 records for each driver. This closes the tested repeated-payload case on Docker 29.1.3. It does not close the deeper case in which a driver or platform exposes two legitimate records with the same timestamp, stream, bytes, and physical generation; such a collision would silently collapse under the current key.

5.8 Recovery work

Direct instrumentation counted records returned by finite Docker rereads. Across ten trials per condition, startup reread amplification was 1.29–1.70 $\times$ (median 1.53 $\times$), and collector restart was 1.23–1.65 $\times$ (median 1.43 $\times$). Exit burst required one full reread in 9/10 trials and two in 1/10, for a median of 1.00 $\times$ and maximum of 2.00 $\times$. All 51 target recovery passes succeeded.

Amplification is target records returned by finite rereads divided by final source-retained unique records. It is not database-write or byte amplification. The 2.00 $\times$ outlier demonstrates why exact insertion is load-bearing even when recovery is otherwise complete. We omit the earlier deployed CPU comparison because LogDeck included an HTTP control plane, parsing, and SQLite while the comparator used a file sink; that measurement did not isolate the recovery mechanism.

6 Related Work

Reliable collection and durable transport. Chukwa is the closest conceptual foundation: agents checkpoint source state and retransmit retained source-local records, while downstream duplicate removal makes replay safe [13]. Kafka provides durable consumer replay [10], and MillWheel combines persistent state with fault-tolerant stream processing [1].

These systems preclude a novelty claim around replay or deduplication. Our contribution is the Docker Logs API boundary, lifecycle suite, attachment race, and exact oracle.

Container log pipelines. Prior systems assemble cluster logging from containerized open-source components [15] or Docker pipelines around the Elasticsearch, Logstash, and Kibana stack [11]. Current collector benchmarks include throughput, resources, rotation, discovery, and missing-log tests [18]. Our study complements them with generation-aware multiplicity and controlled lifecycle boundaries.

File tailing and exporter resilience. Fluent Bit persists file offsets and documents rotation behavior [8]. OpenTelemetry Collector supports queues, retry, and a persistent write-ahead log for exporter resilience [12]. These mechanisms address file replay or downstream delivery; neither by itself guarantees that a Docker API source discovered an ephemeral container. The Alloy result demonstrates the difference within a supported API collector that already persists positions.

Storage and complete capture. Flight Data Recorder studies always-on capture of persistent-state interactions [17], while Kollect targets efficient lossless Windows kernel-event collection [2]. The compressed log processor (CLP) addresses lossless compression and search after collection [14]. This paper studies the upstream question of whether intended records reached durable collector storage at all.

7 Scope and Threats to Validity

Authorship and implementation bias. The author built LogDeck and evaluates it. The study mitigates, but cannot remove, this conflict with an external black-box harness, pinned images, complete expected sets, preserved source observations, fixed-revision ablations, and positive controls in which Alloy and Fluent Bit match LogDeck. The strongest claim is a boundary result, not a universal collector ranking.

External validity. The new supported-API matrix covers one Docker 29.1.3 Linux/arm64 environment. The daemon matrix uses an isolated Docker 29.7.0 engine on the same kernel. Earlier fixed-revision evidence covers independent Ubuntu/arm64 hardware and Docker 24.0.9, but not the new Alloy configuration. Results may differ across Docker versions, logging drivers, kernels, platforms, collector configurations, and large fleets.

Fault and workload coverage. The suite covers attachment, rapid exit, collector restart/crash/pause, same-ID restart, daemon restart with live restore, rotation, two-container concurrency, stream interleaving, and container removal. It does not cover host power loss, filesystem corrup-

tion, long outages that exhaust Docker’s live-restore FIFO, Kubernetes runtime interfaces, arbitrary multiline parsers, or adversarially randomized schedules.

Identity. Exact insertion depends on physical generation, timestamp, stream, and bytes. The two-driver repeated-payload experiment observed zero collisions in 200,000 records but is not a proof of timestamp injectivity. A collision is a silent loss mode and remains an explicit contract exclusion.

Comparator configuration. Alloy 1.18.0 uses its documented Docker source, persistent data path, 100 ms refresh, label-filtered discovery, and a local echo receiver. Other target construction or retention configurations may behave differently. The Fluent Bit control changes both source interface and privilege. The appendix’s Vector control is documented as best effort and is not evidence about Vector as a whole.

8 Artifact and Reproducibility

The ancillary archive contains the standalone Go harness, exact collector configurations, the fixed-revision ablation patch and runner, JSON reports, paper-facing summaries, environment and image provenance, verification scripts, and cryptographic checksum manifests. The daemon runner creates a labeled, privileged Docker-in-Docker engine, verifies live restore, imports the frozen LogDeck image, restarts only the inner daemon, and removes the isolated engine afterward.

The principal new reports contain 120 supported-API collector-runs, 20 daemon collector-runs, 20 removal collector-runs, and 20 observable-identity collector-runs. The archive’s verification script checks denominators, fault-boundary equality, exactness, missing ranges, and duplicate multiplicity. Large raw databases and logs are retained separately; compact reports, manifests, harness code, and provenance accompany the paper.

9 Conclusion

Source-bounded exactness turns a vague losslessness claim into a falsifiable contract: every record still readable from Docker must appear exactly once, and records outside the source horizon are reported separately. The multiset oracle found a real start-to-attachment race, verified its repair, and exposed simultaneous omission and replay that totals conceal.

In the supported-API matrix, LogDeck was exact in 60/60 collector-runs and Alloy in 20/60. Alloy’s successes under startup, pause, and daemon restart make the negative result specific: the tested weakness appears when lifecycle recovery requires reacquiring a source that has already exited or restarted. A private-file Fluent Bit control also recovered

exactly, at the cost of coupling to Docker’s internal storage. When the source container was removed, neither collector recovered the suffix. Together these results identify the useful claim: finite API reconciliation plus exact insertion can recover retained, distinguishable Docker records across the tested lifecycle faults, but no collector can outlive the source or distinguish information Docker does not expose.

A Best-Effort API Control

An earlier independent-cloud matrix compared the fixed LogDeck revision with pinned Vector 0.55.0 across six scenarios and ten trials per collector. LogDeck was exact in 60/60 collector-runs; the configured Vector Docker source was inexact in 60/60, principally through final-record duplication. Focused size curves and raw JavaScript Object Notation Lines (JSONL) localized those duplicates to repeated byte-equivalent events. Vector documents this source as best effort [16]; consequently we treat it as a harness sensitivity control rather than the paper’s main comparator.

B Forward Transport Control

A separate experiment changed the source interface to Docker’s `fluentd` driver and a native Fluent Bit Forward input. When the producer remained alive through collector restart, output matched all 50,000 oracle identities in 10/10 collector-runs. When the producer exited before reconnect, every one of 27 collector-runs retained only its pre-fault prefix, independent of tested asynchronous buffer limits from 128 to 1,048,576 events.

This result concerns transport lifetime rather than Logs-API acquisition. Docker’s in-memory driver queue can bridge a collector restart for a continuing producer, but not producer/logger destruction before drain [4, 7].

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